Impilo vNext: National Health Operating System
Architectural Definition & Implementation Standard
Version 1.2 (Rewritten Canonical — Production-Grade)

## 0. Purpose of This Rewrite

This version eliminates contradictions and production-failure gaps in v1.0 by explicitly defining:

1. Clinical safety vs eventual consistency (bounded staleness, action classes, break-glass, decision logging)
2. Federation for pods (authority boundaries, conflict resolution, revocation propagation, merges)
3. Event strategy that scales (delta-first, snapshots, schema registry, replay/backfill)
4. Operational architecture (DR, RPO/RTO, audit, KMS/HSM, SLOs, release strategy)
5. Workload partitioning (clinical vs telemetry vs analytics buses)
6. API lifecycle + ecosystem governance (dual-mode is real, not a handwave)
7. Real-time payments, claims switching, remittance, and settlement orchestration
8. AI, decision intelligence, and AI insights governance as first-class platform capability

## 1. Executive Summary

Impilo vNext is a Sovereign National Health Operating System (HOS): the national digital spine for health services, identity, shared clinical memory, financing, interoperability, and governance—operating at population scale (15M+) across public, private, and security-sensitive providers.

Impilo vNext is built on a Federated Sovereignty Model:

1. A National Spine provides authoritative national services and policies.
2. Sovereign Pods provide isolated deployments for entities requiring physical/logical isolation (military, major private groups, research) while

maintaining controlled national linkage through a defined federation protocol.

Impilo vNext follows a Ring-Fenced Build Strategy (Ring 0 → Ring 1 → Ring 2) and a Safety-First Consistency Model that makes explicit which clinical actions require synchronous truth, which are projection-safe, and how offline care is safely enabled.

Impilo vNext is also designed as a reusable public-health and operational platform. Public-health, local-authority, and oversight capabilities shall therefore be implemented as shared platform services configured through jurisdiction packs, role scopes, workflow variants, and canonical operational domain objects rather than through duplicate standalone systems. This ensures that city health departments, rural district councils, provincial teams, and national public-health programs can use the same core capabilities without fragmentation or redundant architecture.

Impilo vNext shall also natively support health financing, medical insurance, coverage, preauthorization, claims, contributions, reimbursement, remittance, and settlement as a shared national capability rather than as a disconnected insurer silo. Coverage, entitlement, payer, purchaser, and medical-aid workflows shall consume the same sovereign registries, clinical events, rules, trust controls, payment rails, and analytics foundations already used by care delivery.

Impilo vNext shall also support governed AI and decision intelligence across clinical, public-health, operational, and financing contexts, with strong auditability, human oversight, model governance, and explicit safety boundaries for any AI-assisted action.

## 2. Core Philosophy & System Mental Model

The system is stratified into seven planes, each with a strict mandate:

### 2.1 Kernel Plane (Trust & Truth)

Defines national truth primitives: identity, consent, registries, semantic governance, pricing, finance switching, settlement switching, longitudinal record primitives, and entitlement/capability anchors.

No plane bypasses Kernel policies.

### 2.2 Clinical Plane (Care Execution)

Safety-critical workflows: encounters, orders, results, referrals, telemedicine, inpatient management.

Clinical execution is allowed to proceed under defined bounded staleness rules.

### 2.3 Public Health, Data & Governance Plane (Observation, Operations & Stewardship)

Analytics, surveillance, reporting, research, inspections, complaints, outbreak operations, emergency coordination, governance, and data access controls.

This plane supports both analytical and operational public-health functions.

Never blocks care execution.

### 2.4 Coverage, Financing & Payer Operations Plane

Coverage, eligibility, membership, benefits, preauthorization, utilization management, claims, contributions, provider contracting, reimbursements, remittance, settlement, and purchaser/payer operations.

This plane is native to the HOS and is not a disconnected sidecar insurer stack.

### 2.5 Supply Plane (Logistics & Assets)

Medicines, inventory, procurement, dispatch, devices, equipment, asset registry.

## 2.6 Intelligence, Automation & AI Plane

Governed analytics, predictive models, summarization, anomaly detection, optimization, recommendation support, and AI-assisted insights across clinical, public-health, operational, and financing workflows.

This plane may inform and assist other planes, but it may not silently override Kernel policy, clinical safety, legal entitlement, or financial truth.

## 2.7 Experience Plane (One UI)

Unified “One Experience” framework (3-zone) and surface apps.

## 2.8 Public Health Reuse Principle

Public-health capabilities in Impilo vNext shall be built once as reusable platform services and configured many times for different jurisdictions, programs, and authority levels.

No local-authority, provincial, or national public-health function shall be implemented as a separate standalone product if it can be expressed through shared domain objects, configurable workflows, rules, roles, or templates.

City Health functionality must therefore be treated as the first major configured instance of a broader Public Health Operations capability layer, not as a disconnected municipal application.

## 2.9 Financing Reuse Principle

Coverage, medical-insurance, and payer functions in Impilo vNext shall consume shared sovereign truth and emit interoperable events rather than recreating isolated payer master-data silos.

Identity, provider, facility, terminology, encounter, pricing, audit, trust, payment rails, and analytics foundations shall be shared wherever lawful and semantically appropriate.

## 2.10 AI Governance Principle

AI in Impilo vNext is assistive by default, governed always, and autonomous only by explicit approval in tightly defined low-risk domains. No AI function may silently finalize high-risk clinical, public-health enforcement, entitlement, claims, or settlement decisions without approved human-governed controls.

# 3. Non-Negotiable Architecture Laws (Guardrails v1.2)

These invariants MUST be enforced by design, code, CI policy checks, and runtime controls.

## Law 1 — Canonical Identity & Privacy Separation

- CRID (Client Registry ID): links to PII. Strict governance.
- CPID (Clinical Patient ID): pseudonymous identifier used in clinical workflows and SHR references.
- PRID / FRID: immutable identifiers for providers and facilities.
- INDAWO Site ID: immutable identifier for premises and public-health operational sites.

Cryptographic model is mandatory:

- CPIDs derived via keyed pseudonymization (HSM/KMS-backed keys; rotation supported; controlled re-derivation rules).
- Cross-tenant correlation controls (no stable identifiers leaked across pods unless explicitly permitted by policy).

Coverage, membership, claims, and settlement objects must reference shared sovereign identity truth rather than recreating isolated person/provider/facility masters.

## Law 2 — One Enforcement Plane (Central Policy Decision)

Authentication/authorization is centralized:

- Edge/API Gateway + Service Mesh enforcement (Envoy)
- Policy engine (OPA)

Services DO NOT implement bespoke auth logic beyond local authorization checks based on verified identity context.

Step-up auth and risk controls for sensitive actions are enforced centrally.

Law 2A — Logical Service Boundaries Are Not Automatic Deployment Boundaries

Named services and components in this architecture are logical service boundaries first. They do not automatically imply separate repositories, clusters, deployables, or operational teams from day one.

Default rule: start as modular components within the ring-appropriate execution boundary unless there is a proven extraction trigger.

Acceptable extraction triggers include:

1. materially different scaling profile
2. materially different resilience requirement
3. materially different data protection boundary
4. materially different team ownership
5. demonstrated operational risk if left in-process

This law exists to prevent premature microservice sprawl while preserving the integrity of the canonical component model.

Law 3 — Event Schemas Are First-Class Artifacts

All domain events must:

- use a versioned schema registered in a Schema Registry
- include:
  - event_id, event_type, schema_version
  - correlation_id, causation_id
  - idempotency_key

  - producer, tenant_id/pod_id
  - subject_id (entity id) and subject_type
  - occurred_at, emitted_at
- comply with compatibility policy: backward compatible by default unless explicitly breaking with migration plan

Financial and settlement events must additionally define their financial reference, amount, currency, and authoritative lifecycle state where relevant.

AI-linked workflow events must additionally define model identifier/version, inference class, and human-review requirement where relevant.

Law 4 — Delta-First Events + Snapshot Strategy

Kernel emits delta events (create/update/delete with changed fields) as default.

Kernel emits periodic snapshots and on-demand snapshot endpoints for:

1. new consumers
2. recovery
3. backfill
4. coverage truth refresh
5. settlement and financial verification refresh
6. AI feature rebuild where approved

“Full state in every event” is prohibited for large domains (terminology, catalogs, privileges).

Law 5 — Projection Strategy With Bounded Staleness

FHIR is the canonical clinical write model for longitudinal record primitives (Butano SHR).

Read models may be Elastic/SQL/OLAP—BUT every clinical action must declare its Consistency Class (see Law 6).

Projection consumers must implement:

1. ordering strategy (partition keys)
2. replay handling
3. poison message strategy
4. backfill strategy
5. staleness reporting

Financial and settlement projections must also define correctness thresholds, settlement finality expectations, and reconciliation fallback behavior.

AI feature and inference consumers must also define model compatibility, drift sensitivity, replay use, and deactivation path.

Law 6 — Clinical Safety Consistency Classes

Every clinical capability MUST be categorized and enforced:

Class A: Hard-Truth Required (Sync Check)

- controlled substances prescribing
- high-risk procedures
- clinician privilege checks
- consent revocation-sensitive actions
- billing/claims submission finalization where clinical truth is needed

Requires synchronous validation against Kernel truth (or signed entitlement + freshness proof).

Class B: Bounded-Stale Allowed

- routine documentation, vitals, low-risk notes
- orders entry (non-controlled) with later reconciliation

Allowed to proceed on projections if:

1. staleness ≤ defined threshold (e.g., 5–30 min by domain)

2. action is logged with decision evidence and reconciliation path

Class C: Always Allowed Offline (with Entitlement)

- emergency care capture
- triage, vitals, basic notes

Requires signed offline entitlements, strict audit logging, and post-sync reconciliation.

Law 6A — Financial Truth & Settlement Classes

Every coverage, payer, claims, and settlement capability MUST declare its financial execution class.

Finance Class F1: Immediate Hard Truth Required

- real-time eligibility for service commitment
- point-of-service guarantee of payment
- funds reservation
- final claim approval for instant settlement
- settlement confirmation
- reversal and refund authorization

Requires authoritative synchronous validation.

Finance Class F2: Near-Real-Time Allowed With Verified State

- preauthorization triage
- provisional claim adjudication
- low-risk provider quote validation
- remittance preparation

Allowed with bounded lag and automatic re-check before irreversible financial action.

Finance Class F3: Deferred or Batch Execution Allowed

- end-of-day provider payment batching
- contribution invoicing
- actuarial computation
- reserve tracking
- reconciliation batch runs

Allowed in asynchronous workflows.

Law 6B — AI Decision Classes

Every AI capability MUST be categorized.

AI Class I1: Insight Only

- dashboards
- trend surfacing
- summarization
- anomaly hints

No direct operational effect.

AI Class I2: Recommendation With Human Action

- claim triage suggestion
- risk score
- preauthorization prioritization
- coding suggestion
- outbreak hotspot recommendation

Human acceptance is required.

AI Class I3: Governed Automation in Low-Risk Domains

- queue routing
- reminder prioritization
- low-risk document classification
- duplicate claim pre-screening

Permitted only with explicit approval, monitoring, rollback path, and audit.

AI may not silently finalize high-risk clinical decisions, high-risk public-health enforcement actions, membership termination, final claim denial in sensitive categories, or final settlement release unless explicitly approved by policy and governance.

Law 7 — Offline Is a Formal Policy Matrix (Not a Feature)

Offline support requires:

- explicit per-action offline allowance
- signed entitlements issued by Trust Layer:
  - scope (actions), time window, device binding
  - patient context constraints (optional)
- conflict resolution rules per domain
- mandatory audit events for offline actions
- financial and AI restrictions in offline mode where applicable

Law 8 — Workload Partitioning (Separate Buses)

- Clinical Bus: safety-critical domain events (low latency, high reliability, bounded payloads).
- Public Health Operations Bus: surveillance, inspection, complaint, incident, campaign, and response events.
- Financial & Settlement Bus: coverage, claims, payment authorization, remittance, and settlement events.
- Telemetry/IoT Bus: high-throughput device events (separate partitions, retention policies).
- Analytics Bus / Lake ingestion: bulk pipelines, ETL, near-real-time analytics, and AI feature ingestion.

Mixing these is prohibited.

Law 9 — Service Count Discipline (Module-First, Platform-Backed)

Start as modular monolith per ring unless there is a proven need to extract.

“Sovereign Kernel services” are allowed only where they are truly shared DPI primitives and must adhere to strict operational standards (SLOs, versioning, backwards compatibility).

Law 10 — Real-Time Financial Actions Must Be Idempotent and Reversible by Design

Any authorization, reservation, remittance, or settlement-triggering action must accept idempotency keys, support duplicate suppression, define reversal or compensation semantics, define a reconciliation pathway, emit auditable events, and support dispute and exception handling.

Law 11 — AI Must Be Auditable, Versioned, and Revocable

Any AI-generated output used in workflow must record model identifier, model version, input context class, confidence or score where relevant, human acceptance or override where relevant, decision outcome linkage, and timestamp/actor context.

4. Sovereign Kernel (Ring 0): Authoritative DPI Services

Ring 0 services define national truth primitives, financial rails, AI governance control points, and federation controls. They have zero dependencies on Ring 1+.

Ring 0 Mandatory Services (Canonical)

TSHEPO — Trust & Policy Service

- IAM (OIDC), consent evaluation, token issuance
- device identity & attestation
- offline entitlement issuance (signed JWT/CBOR tokens)
- step-up auth integration and risk scoring hooks
- audit decision logging (policy decisions)

VITO — Client Registry (MPI)

- CRID management, dedup/merge

- CPID mapping rules and re-keying under governance
- linkage to Ubomi and national identifiers under policy
- beneficiary linkage foundations for coverage and payer domains

VARAPI — Provider Registry

- practitioner identity, licensure, privileges, role assignments
- privilege revocation propagation rules (federation-aware)

TUSO — Facility Registry

- facility topology, departments, capabilities, service availability
- facility identity federation for pods
- canonical registry for health service-delivery facilities

INDAWO — Site & Premises Registry

- regulated premises and public-health operational sites
- canonical identifiers, geospatial and jurisdictional linkage, lifecycle status
- operator/owner references where relevant
- inspection, compliance, surveillance, complaints, and emergency-response linkage hooks

TUSO governs clinical and service-delivery facilities.

INDAWO governs non-clinical regulated premises and public-health operational sites whose semantics, lifecycle, and governance do not fit the facility model.

MSIKA — Product, Service, Tariff & Benefit Catalog Registry

- orderables, billables, tariffs
- benefit baskets and reimbursement references
- versioned catalogs and effective date management
- delta + snapshot emission

ZIBO — Terminology & Semantic Governance

- ICD-11, SNOMED, LOINC mapping governance
- value sets, concept maps, version lifecycles
- national governance workflows (approvals, publishing)
- claims coding alignment and AI-safe semantic references

BUTANO — Shared Health Record Core (FHIR Longitudinal Memory)

- canonical longitudinal record primitives
- FHIR profile governance + validation
- subscription hooks (where permitted)

UBOMI — CRVS Interface

- births/deaths linkage protocols
- identity event handling with reconciliation patterns

MUSHEX — Finance Engine / Claims Switch / Settlement Rail

- claims processing, billing, remittance, and settlement switching
- real-time authorization, reservation, guarantee-of-payment, and payout orchestration
- consumes Msika pricing and national coverage rules
- integrates fraud signals, reconciliation, and audit requirements

Ring 0 Additional Mandatory Platform Primitives (Often Forgotten)

These are not "later":

1. Schema Registry + Contract Testing Harness
2. Audit Ledger Service (Tamper-Evident)
3. Key Management & Secrets Service (KMS/HSM integration)
4. Developer Platform Portal (SDKs, sandbox, API keys, docs, deprecation)
5. AI Governance & Model Registry Control Plane

If you don't build these, you will not operate dual-mode services safely.

5. Federation & Sovereignty Strategy (Pod Model v1.2)

Federation is not optional. Pods are not forks. Pods are governed.

## 5.1 Deployment Levels

### Level 1 — National Spine (Shared National Tenant)

Serves most public facilities.

Uses logical tenant isolation; high-risk domains may use crypto-segmentation and dedicated partitions.

RLS alone is not sufficient—must be layered with policy enforcement + encryption + rigorous testing.

### Level 2 — Sovereign Pod (Isolated Deployment)

Full stack isolated instance for entities requiring separation.

Connects to National Spine using Federation Protocol APIs and event channels.

Must declare:

1. which domains it is authoritative for locally
2. what it consumes from National Spine
3. what it publishes back (reporting obligations, identity links)
4. what financial/settlement participation model it follows where relevant
5. what AI participation limits apply where relevant

### Level 3 — Export/Country Kit

Configuration-injected localization (IDs, phone formats, currency, locale, regulatory profiles).

Separation of code vs configuration enforced.

## 5.1A Jurisdiction Profiles / Operational Packs

Not every operational variation in the ecosystem requires a sovereign pod. In addition to the National Spine, Sovereign Pod, and Export/Country Kit models, Impilo vNext shall support jurisdictional configuration packs for local-authority and public-sector operational variation within the same national platform.

Examples include:

- City Health Pack
- Rural District Council Health Pack
- Provincial Public Health Oversight Pack
- National Public Health Oversight Pack
- Port Health Pack
- School Health Pack
- Medical Aid / Scheme Operations Pack
- Employer Health Scheme Pack
- Purchaser / Payer Oversight Pack

A jurisdiction pack may define:

1. enabled modules and menu surfaces
2. geography scope and hierarchy bindings
3. workflow variants
4. role and approval matrices
5. checklist and form libraries
6. dashboard definitions
7. document templates
8. reporting obligations
9. legal and regulatory references

Jurisdiction packs are configuration artifacts, not code forks. Their purpose is to maximize reuse while allowing lawful and operationally necessary variation.

5.2 Federation Protocol (Mandatory)

Federation defines Authority, Linkage, and Conflict Resolution.

A. Authority Boundaries (Per Domain)

Each pod must be one of:

1. National-Authoritative Consumer (reads national truth; may have local copies)
2. Pod-Authoritative with National Linkage (local truth, national references)

Dual-Authority prohibited unless explicit conflict rules exist (avoid).

Default rule:

Identity (CRID/CPID), terminology, facility master IDs, site/premises master IDs, provider licensure, and core settlement/payment rail truth are National-Authoritative unless legally impossible or unless an explicit pod-authoritative-with-linkage rule has been approved for the domain.

B. Linkage, Merge, Revocation Propagation

- National merges in Vito emit Merge Events with mapping, not destructive overwrite.
- Pods must implement merge reconciliation within bounded time.
- Consent revocation and privilege revocation propagate via High-Priority Control Channel (not best-effort).

C. Patient Identity Across Pods

Cross-pod identity linkage is explicitly consent-governed.

Default: pods hold local clinical identifiers that map to national CPID via approved linkage. No silent correlation.

D. Reporting Obligations

National surveillance/statutory reporting is a policy requirement:

- pods must publish mandated aggregates/events under legal constraints

Data Access Governance controls exports and purpose limitation.

Where pods originate claims, provider payments, or scheme operations but do not hold settlement authority, they must hand off settlement orchestration to the national MusheX rail through defined contracts and events.

## 6. Clinical Safety Model (Production Rules)

### 6.1 Truth Evaluation Strategy

Class A actions require one of:

1. synchronous Kernel policy decision, OR
2. offline entitlement with validity + freshness proof + later reconciliation rules

F1 financial actions require synchronous authoritative validation unless governed signed proof exists.

### 6.2 Break-Glass (Mandatory)

Break-glass allows care continuation when policy checks fail or are unavailable.

Requires:

1. explicit justification capture
2. time-bounded elevated access
3. mandatory audit + review workflow
4. alerting to compliance

Break-glass does not automatically authorize irreversible financial release.

### 6.3 Decision Evidence Logging

Every policy evaluation (allow/deny/break-glass) must emit an immutable audit event containing:

- actor, patient reference, action, decision, reason codes, policy version, context

This is how you survive inquiries.

## 6.4 Real-Time Financial Clearance Model

Impilo vNext shall support real-time financial clearance at the point of service where applicable. Financial clearance may include eligibility verified, contracted provider verified, benefit and package checked, exclusion and waiting-period checked, preauthorization requirement determined, member liability computed, payer liability estimated or approved, and guarantee-of-payment or reservation token issued where appropriate.

## 6.5 Settlement State Model

At minimum the architecture must support these financial states:

1. initiated
2. eligibility-verified
3. preauthorized
4. reserved
5. provisionally-adjudicated
6. approved
7. remitted
8. paid
9. settled
10. reversed
11. reconciled
12. disputed
13. recovered

## 6.6 AI Safety Model

AI outputs may be informational, recommendatory, or governed low-risk automation, but never ungoverned silent high-risk automation. AI may recommend, prioritize, summarize, classify, score, and detect anomalies, but must remain subject to explicit governance, audit, and human oversight where required.

# 7. Data, Events, and Read Models

## 7.1 Event Backbones

- Clinical Backbone: reliable, ordered per entity, short retention + archive.
- Public Health Operations Backbone: reliable operational events for surveillance, inspections, outbreaks, campaigns, complaints, and response.
- Financial & Settlement Backbone: coverage, claims, authorization, remittance, settlement, reversal, and reconciliation events.
- Telemetry Backbone: high throughput, long retention options, separate scaling.
- Analytics Backbone: stream processing into lake/warehouse and governed AI feature ingestion.

## 7.2 Projection Requirements

Every projection must have:

1. bootstrap method (snapshot + catch-up)
2. staleness monitor and alerting
3. replay-safe idempotency
4. schema compatibility validation
5. bounded lag SLAs per domain

Financial projections must additionally define correctness guarantees, reconciliation path, and finality handling.

AI feature and inference projections must additionally define model compatibility and drift-handling expectations.

## 7.3 FHIR as Record Truth (Butano)

Butano stores canonical longitudinal record resources.

The Clinical Plane may store encounter-execution state locally but must link to Butano for longitudinal memory.

FHIR profiles and validation rules are governed via Zibo + national governance workflow.

## 7.4 Canonical Operational Domain Model (Beyond Clinical Care)

Impilo vNext must define and govern canonical domain objects not only for clinical care and registries, but also for public-health and local-authority operations. These objects are national platform primitives and must not emerge as isolated app-specific models.

Minimum shared operational objects include:

- Premises / Regulated Site
- Household / Community Unit
- Geographic Area / Ward / Zone / Catchment
- Public Health Event
- Complaint / Alert / Signal
- Inspection
- Investigation
- Finding / Violation
- Notice / Enforcement Action
- Sample / Environmental Test
- Outbreak / Incident
- Campaign / Outreach Session
- Response Task / Deployment
- Public Communication Activity

These objects may be profiled differently by jurisdiction or program but must share canonical identifiers, lifecycle states, auditability, and interoperability contracts.

## 7.5 Canonical Coverage, Financing & Payer Domain Model

Minimum shared financing objects include:

- Scheme
- Plan / Package
- Membership
- Beneficiary Enrollment
- Coverage Period
- Benefit Rule
- Waiting Period Rule
- Exclusion Rule
- Co-Pay / Deductible / Limit Rule
- Provider Contract
- Tariff Schedule
- Eligibility Decision
- Preauthorization
- Utilization Review Case
- Claim
- Claim Line
- Adjudication Decision
- Appeal
- Contribution Schedule
- Invoice
- Receipt
- Payment Reservation
- Guarantee of Payment
- Provider Payment Batch
- Remittance Advice
- Settlement Instruction
- Reconciliation Item
- Recovery / Clawback
- Fraud / Integrity Case

These objects must reference shared sovereign truth wherever possible rather than inventing isolated payer copies.

7.6 AI & Insight Objects

Minimum governed AI-related objects include:

- Model

- Model Version
- Feature View
- Inference Record
- Recommendation Record
- Human Override Record
- AI Alert
- Drift Event
- Model Approval / Withdrawal Record

## 8. Operational Architecture (SRE is part of architecture)

### 8.1 Reliability Targets

Each Ring 0 service must define:

1. SLOs (availability, latency, freshness for projections)
2. error budgets
3. capacity assumptions
4. settlement finality expectations where relevant
5. AI inference and model-governance latency/availability expectations where relevant

### 8.2 Disaster Recovery

Mandatory DR posture:

1. defined RPO/RTO per ring and service class
2. automated backups
3. restore drills
4. runbooks and incident escalation paths
5. payment reconciliation continuity
6. claims and settlement recovery path
7. AI model registry recovery and safe degradation mode

### 8.3 Release Governance

- progressive delivery: canary + feature flags
- strict API versioning + deprecation windows

- schema compatibility gates in CI
- financial control gates for payment/settlement-impacting changes
- AI safety gates for model and workflow-impacting changes
- ring-based release trains (Kernel is slow and stable; outer rings can move faster)

8.4 Security Hardening

- KMS/HSM-backed keys for identity pseudonymization and signing
- secrets rotation
- mTLS
- PAM for admin access
- security event pipeline into SIEM
- rate limiting, quota, abuse controls at gateway level
- financial rail protection controls
- AI inference gateway protection controls

9. Dual-Mode National Shared Services (Made Real)

Dual-mode means:

Kernel services support:

1. native Impilo apps
2. external third-party apps via stable APIs
3. sovereign pods
4. medical-aid and payer participants
5. approved AI and analytics consumers

Therefore mandatory:

1. developer portal, SDKs, sandbox, mock servers
2. API keys / client registration
3. contract tests for partners
4. versioning and deprecation policy
5. onboarding certification process
6. coverage/payer participant integration profile
7. payment/settlement participant integration profile

8. AI and analytics consumer integration profile

If this isn’t built, “dual-mode” is fiction.

10. Ring-Fenced Build Strategy (Corrected)

Ring 0 — Kernel + Platform Primitives (Do Not Skip)

Tshepo, Vito, Varapi, Tuso, Indawo, Msika, Zibo, Butano, Ubomi, MusheX

plus:

1. Schema Registry + Contract Test Harness
2. Audit Ledger
3. KMS/HSM integration
4. Gateway + OPA enforcement
5. Observability baseline (metrics/logs/traces)
6. Developer portal baseline
7. AI governance/model registry baseline

Exit criteria:

1. policy decisions logged and auditable
2. snapshot + delta functioning
3. federation channels defined
4. DR drills demonstrated for Kernel
5. financial authorization and settlement rail baseline proven
6. AI governance baseline operational

Ring 1 — Steel Thread Clinical + Coverage + Finance (Minimal Viable National Care)

- Patient Care Tracker (PCT)
- Orders/Results Orchestration
- Referral & Care Network
- Scheduling/Capacity
- Basic inpatient (bed management minimal)
- Eligibility and Coverage Service

- Membership and Beneficiary Administration
- Benefit Rules and Product Administration
- Preauthorization and Utilization Management
- Claims Capture and Adjudication
- Contribution, Billing and Collections
- Provider Payments and Settlement Operations
- Member and Provider Self-Service Baseline
- Finance flows that prove MusheX + Msika integration

Exit criteria:

1. Class A/B/C enforcement working
2. Finance Class F1/F2/F3 enforcement working
3. break-glass operational
4. offline entitlements tested
5. end-to-end “care → record → eligibility → claim → settlement” verified

Ring 1 Minimum Dependency Clarification

Although the broader workflow, rules, and content platforms mature in Ring 2, Ring 1 is permitted and expected to depend on a minimal baseline of workflow orchestration, forms capability, notification support, and AI-assist capability where required to complete the national steel thread safely.

Ring sequencing defines maturity priority, not an absolute prohibition on minimal enabling dependencies.

Ring 2 — Public Health, Scale, Intelligence & Governance Layer

- public-health operations, surveillance, and outbreak tooling
- analytics, reporting, and governance platforms
- content, workflow, and rules platforms beyond Ring 1 minimums
- IoT, inventory, and asset ecosystems
- data governance tooling for research and exports
- fraud, waste, abuse, and integrity controls
- advanced settlement and purchaser intelligence
- AI and decision-intelligence services at scale

Exit criteria:

1. no impact on Ring 1 latency/safety
2. data governance controls enforceable
3. telemetry separated and scalable

11. Component Model (Reframed for Production)

You can still have a large feature list, but the architecture must enforce what is shared, what is local, and what is optional.

11.1 Planes and Components (Canonical)

I. Kernel Plane (Ring 0)

- Tshepo, Vito, Varapi, Tuso, Indawo, Msika, Zibo, Butano, Ubomi, MusheX
- Audit Ledger (tamper-evident)
- Schema Registry + Contracts
- KMS/HSM integration
- API Gateway + OPA enforcement
- Federation Control Service (policies + routing + obligations)
- AI Governance & Model Registry Control

II. Clinical Plane (Ring 1)

- Patient Care Tracker / Encounter Execution
- Orders & Results Orchestration
- Referral / Care Network
- Scheduling & Capacity
- Bed & Critical Resource Management (minimum viable subset)
- Telemedicine Execution Framework
- Departmental clinical modules (as plugins/modules, not 20 microservices)

III. Coverage, Financing & Payer Plane (Ring 1)

- Scheme and Product Administration
- Membership and Beneficiary Administration

- Provider Contracting and Network Management
- Eligibility and Entitlement Service
- Preauthorization and Utilization Management
- Claims Capture and Adjudication
- Contribution Billing and Collections
- Provider Payment, Remittance and Settlement Operations
- Member Portal
- Provider Portal
- Complaints, Appeals and Query Management

IV. Integration, Rules & Content Plane (Ring 2)

- Interoperability Gateway (FHIR/other standards)
- Rules & Workflow Engine (BPMN/DMN)
- Form & Clinical Content platform
- Search service (clinical safe subset vs global search)
- Notifications hub
- Identity assurance/risk engine (integrated with Tshepo)

V. Public Health, Data, Analytics & Governance Plane (Ring 2)

- data pipeline, NDR, BI/reporting
- surveillance / eIDSR
- outbreak and incident operations
- inspections, complaints, and field operations
- campaign and outreach orchestration
- data access governance enforcement (not just a portal)
- fraud/waste/abuse analytics
- payer and utilization intelligence

VI. Supply Plane (Ring 2)

- eLMIS/inventory/procurement
- device/IoT integration (separate bus)
- asset/equipment registry

VII. Intelligence, Automation & AI Plane (Ring 2)

- model registry
- governed feature views
- inference gateway
- summarization and coding assist
- anomaly detection
- fraud and integrity scoring
- triage and prioritization engines
- forecasting and optimization models
- drift monitoring
- explanation and override layer

VIII. Experience Plane

- One UI / One Experience (3-zone)
- facility ops console
- payer and scheme operations console
- citizen portal & wellness suite
- provider mobile app
- AI-assisted workbench surfaces

IX. Execution & Resilience Frameworks

- offline sync & edge execution framework
- observability
- security hardening
- support/helpdesk/knowledge

## 12. One UI Guidelines (Enforced)

### 12.1 The 3-Zone Framework

All user interaction occurs in:

1. Work (facility operations + clinical execution)
2. My Professional (licensure, CPD, privileges, scheduling personal)
3. My Life (citizen portal, wellness, household care)

### 12.2 Non-Interference Rule (Clarified)

National Services (Vito/Msika/Zibo etc.) can be accessed post-login as modules but must not disrupt active encounter execution.

Enforcement:

1. encounter UI state is isolated
2. kernel module actions that change truth (e.g., merge, tariff updates) must be performed in separate workflow contexts and propagate via events

12.3 Financial Transparency Rule

Where relevant, the UI must surface coverage state, member liability, payer liability, preauthorization status, claim status, remittance status, and dispute or exception flags.

12.4 AI Transparency Rule

Where AI is used, the UI must make clear that AI was used, what output was generated, what confidence or rationale exists where available, and whether human review is still required.

13. Definition of Done for "Production-Ready"

vNext is not production-grade until all are true:

1. Clinical safety classes enforced end-to-end (A/B/C)
2. Break-glass exists, audited, reviewable
3. Audit ledger proves who/what/why for sensitive operations
4. Schema registry + compatibility gates active in CI
5. Snapshot + delta proven with replay and new consumer bootstrap
6. Federation protocol operational with revocation propagation
7. DR drill completed with documented RPO/RTO results
8. Observability provides SLIs/SLOs and paging
9. Key management is HSM/KMS-backed with rotation plan
10. Dual-mode platform has developer portal + versioning/deprecation policy

11. One jurisdiction pack proven live without code fork
12. One sovereign pod tested end-to-end through federation registration, revocation propagation, and merge reconciliation
13. One offline reconciliation drill completed under real low-connectivity conditions
14. One external integration certified through portal, contracts, and event compatibility
15. One public-health operational workflow proven end-to-end (for example inspection, complaint, or outbreak handling)
16. One coverage and claims workflow proven end-to-end
17. One real-time financial clearance and settlement workflow proven end-to-end
18. One reversal/refund/recovery path proven
19. One AI-assisted workflow proven with audit, human override, and rollback
20. One fraud/integrity workflow proven across care and financial events

## 14. Final Architectural Position

Impilo vNext succeeds only if it behaves like national DPI and not "a big app with microservices."

This manifest makes the hard choices explicit: safety, federation, operations, governance, reusable public-health capability, native coverage and medical-insurance operations, real-time financial execution, AI-enabled but human-governed intelligence, and scalable eventing.